

BOȘȚINĂ GEORGE EDUARD (Eduard Bostina)

CS Student | Linux Kernel & Systems Programming

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Third-year Computer Science student at Politehnica University of Bucharest (GPA 9.4/10) contributing to the mainline Linux kernel through Google Summer of Code 2026 with The Linux Foundation. Lead architect of a 64-bit AArch64 OS kernel that boots on real hardware to a userspace shell. **Seeking a Summer 2027 systems/embedded engineering internship; open to relocation across the EU.**

OPEN SOURCE CONTRIBUTIONS

Google Summer of Code 2026 Contributor, Linux Kernel May - Aug 2026
The Linux Foundation (open source organisation) | Remote

- Converted a 50-binding set of legacy Device Tree bindings (mostly Texas Instruments) to YAML dt-schema and updated the dependent .dts sources, enabling automated hardware-description validation across the ARM/OMAP platform tree.
- 22 patches merged into mainline Linux**, including 18 binding conversions across 16 subsystems (devicetree, MFD, clock, pinctrl, MMC, input, RTC, ASoC and others), through multiple rounds of maintainer review. Completed the program in August 2026.
- Verifiable upstream work: lore.kernel.org/all/?q=egbostina@gmail.com

EDUCATION

Politehnica University of Bucharest Oct 2024 - Jul 2028 (expected)
Faculty of Automatic Control and Computer Science
B.Sc. in Computer Science & Engineering (in progress) **GPA: 9.4 / 10**

PROJECTS

- Perspicua** C, AArch64, Assembly 2025 - Present | github.com/perspicua/perspicua
Lead architect and primary contributor on a 4-person 64-bit AArch64 kernel for the Raspberry Pi 4B, written from scratch in C and assembly (~31k LOC). **Boots on real hardware to a userspace shell:** SMP across 4 cores, virtual memory with copy-on-write, 37 POSIX-style syscalls (fork/exec, signals, pipes, mmap), a VFS with FAT32 root, page/block caches, and 15 userspace programs. 19 in-kernel test suites, KASAN, lockdep, and an in-kernel debugger.
- Aegis** C++17, libsodium, zlib 2025 | github.com/edybostina/aegis
Cross-platform CLI for authenticated file encryption (Linux/macOS/Windows) using AEAD (XChaCha20-Poly1305). Passphrase and keyfile modes, optional compression, recursive directory processing, documented security design; 5 tagged releases with CI via GitHub Actions.
- Matrix-lib** C++17, SIMD, Multithreading 2025 | github.com/edybostina/matrix-lib-cpp
Header-only C++ linear-algebra library: 50+ operations, LU/QR decomposition, and eigenvalues. Optimized with SIMD (AVX2/NEON) and automatic multithreading for **4-8x speedups** on large matrices, plus an optional BLAS backend (up to 500x versus a naive implementation).
- Rasterizer** C++, SDL2 2025 | github.com/edybostina/rasterizer
Software 3D renderer with no GPU: vertex transform, frustum clipping, scan-line rasterization, .obj loading, and texture mapping. Renders a 5.9k-triangle scene at 1920x1080 in real time; tile-based multithreading gives a **2.6x speedup**.

TRAINING, AWARDS & CERTIFICATIONS

Google ARM Summer School | Embedded Linux Summer 2025
Selected participant | Politehnica Bucharest | One-week intensive

- Implemented a complete ARM boot chain (TF-A, U-Boot, kernel) and a custom Buildroot Linux distribution (kernel, rootfs, initramfs) for the NXP i.MX93 (Cortex-A55), including TrustZone/TEE secure-world configuration and custom device drivers.

- Awards & Certifications**
- Gold Medalist (Top 1%), National Olympiad in Applied Informatics, AcadNet 2024**
 - Global Finalist (6th worldwide, 2nd in Romania), JetBrains Student Quiz 2025
 - NVIDIA Certificate: Building AI Agents with Multimodal Models (2025)

TECHNICAL SKILLS

Languages	C/C++, Assembly (AArch64, NASM), Python, Bash/Shell, C#, Java
Systems & Tools	Linux, Buildroot, U-Boot, QEMU, GDB, CMake, Git, Docker, GitHub Actions (CI/CD), SDL2, CUDA
Expertise	Device drivers, kernel modules, Device Tree, cross-compilation, POSIX, concurrency, memory management, embedded Linux, systems programming
Spoken	Romanian (native), English (fluent)